TCG-IDS: Robust Network Intrusion Detection via Temporal Contrastive Graph Learning

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***Abstract*—In the era of zero trust security models and next-generation networks (NGN), the primary challenge is that network nodes may be untrusted, even if they have been verified, necessitating continuous validation and scrutiny. Effective intrusion detection systems (IDS) are crucial for continuously monitoring network traffic and identifying potential threats. However, traditional IDS approaches often struggle to keep pace with evolving threats, requiring extensive supervised training on labeled datasets. This limitation leads to high false positive rates, low detection accuracy, and a failure to provide real-time detection, thereby undermining the security of NGNs. This paper proposed the first self-supervised learning-based IDS, designed on temporal contrastive graph neural network (GNN), namely TCG-IDS. It innovatively integrates three contrastive learning strategies: temporal contrasting to capture temporal dependencies, asymmetric contrasting to account for the diverse interactions within network data, and masked contrasting to enhance the learning of node representations by masking parts of the data during training. Performance evaluation was conducted on two publicly available network traffic datasets, NF-CSE-CIC-IDS2018-V2 and NF-UNSW-NB15-V2. TCG-IDS achieved a balanced accuracy of 99.48% and 91.48% on two datasets respectively, significantly outperforming state-of-the-art graph learning models. In multi-class detection, TCG-IDS attained a mean false positive rate of 4.15% and 3.34% on the two datasets respectively. Besides, it exhibits high efficiency with its running time of 0.37s and 0.51s on the two datasets to predict per batch of 100 samples. Results highlight the effectiveness and efficiency of TCG-IDS in accurately detecting various types of network intrusions. This work significantly advances the field of network intrusion detection via self-supervised temporal graph learning, offering a promising solution for future network security systems.**

***Index Terms*—Cyber security, temporal graph neural network, intrusion detection system, network security.**

I. Introduction

THE advent of zero trust security models and NGNs has underscored the need for advanced security measures in modern network environments [1], [2]. Conventional security architectures, which assume implicitly trusted devices once authenticated, are increasingly inadequate and pose significant security risks [1]. IDSs [3], [4], [5], [6], [7] are essential for identifying and mitigating potential threats by monitoring network traffic for signs of malicious activities [8]. However, traditional IDS approaches, such as signature-based and anomaly-based detection, often struggle to keep pace with the evolving threat landscape, resulting in high false positive rates and limited effectiveness against novel attacks [9]. This challenge is particularly acute for NGNs, which demand more advanced security measures. Zero trust security models, emphasizing continuous, context-aware, and dynamic authentication, offer a promising solution by requiring ongoing trust evaluation and leveraging AI and deep learning to enhance intrusion detection and prevention [1]. Addressing these aspects is crucial for securing NGNs, highlighting the importance of research on advanced IDS in this field [10].

A. Motivations

Recently, researchers have explored various learning-based approaches, including feature-based, deep learning-based, and GNN-based methods. Feature-based methods rely on extracting representative traffic features and using machine learning classifiers, showing promise in detecting known attacks but struggling with unseen patterns [11], [12]. Deep learning-based methods, such as CNNs and RNNs, excel in learning complex patterns from high-dimensional data but require extensive labeled data for supervised training [13]. GNN-based methods, which naturally model the graph structure of network traffic, have emerged as a promising approach, with static GNNs like GCNs and GATs learning node representations from static snapshots [14], [15], [16], [17]. However, these fail to capture temporal dynamics. To overcome this, dynamic GNNs incorporate temporal information,

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either through discrete-time models learning from sequences of static snapshots or continuous-time models capturing the ongoing evolution of node representations [18], [19]. Dynamic GNNs, while effective at capturing temporal dependencies, often rely on supervised learning, limiting their adaptability to evolving threats and novel attacks. This dependence on extensive labeled data and predefined signatures restricts their ability to minimize false positives in real-world scenarios.

Despite advancements in machine learning-based IDS, existing approaches face significant challenges [11], [12], [13], [14], [15], [16], [17], [18], [19], [20]. Most focus on static network data, overlooking temporal dynamics and evolutionary patterns critical to understanding network behavior. Additionally, their reliance on supervised learning limits applicability in real-world scenarios where labeled data is scarce or expensive. These methods also exhibit low generalizability, often requiring extensive training data to achieve high detection accuracy. Addressing these limitations necessitates self-supervised learning models for NGNs, leveraging GNNs to capture temporal patterns and enhance detection accuracy in dynamic and evolving network environments.

B. TCG-IDS

To bridge the research gap, TCG-IDS is proposed as the first self-supervised temporal contrastive GNN-based system for intrusion detection that leverages temporal dynamics and spatial dependencies in network traffic data to learn robust representations of network entities and their interactions. These representations are robust because they effectively capture the complex, evolving patterns of normal and malicious behaviors in the network, enabling the model to accurately detect anomalies even in the presence of noisy or incomplete data. It integrates three key contrastive learning strategies: temporal contrasting, asymmetric contrasting, and masked contrasting. Temporal contrasting ensures that the model captures the temporal evolution of network behavior, while asymmetric contrasting focuses on the relationships between nodes, enhancing the model's ability to detect anomalies in node interactions. Masked contrasting helps the model learn from incomplete data, improving its robustness and generalization capabilities. By incorporating these strategies, TCG-IDS can effectively detect novel attacks with minimal false positives, even in scenarios with limited labeled data.

C. TCG-IDS *'s Ability for Constructing Zero-Trust Model*

Zero trust security models require continuous validation of all network entities, emphasizing dynamic, real-time monitoring to detect conflicts or malicious activities. TCG-IDS addresses this by employing self-supervised learning on unlabeled datasets, leveraging temporal and spatial dependencies through a GNN framework. Its use of temporal, asymmetric, and masked contrasting strategies ensures robust node behavior representation, enabling comprehensive monitoring and effective detection of novel attacks with minimal false positives. This capability strengthens zero trust environments, offering a scalable and reliable solution for next-generation network security.

D. Technical Novelty

TCG-IDS addresses key limitations of existing IDS approaches with three innovative contrastive learning strategies. (i) Temporal contrasting captures the temporal evolution of network behavior, overcoming the static data analysis issues in traditional IDS. (ii) Asymmetric contrasting enhances anomaly detection by focusing on node relationships and predicting neighboring embeddings, effectively identifying subtle malicious interactions. (iii) Masked contrasting improves robustness and generalization by enabling learning from incomplete data, tackling the challenge of limited labeled datasets and noisy inputs. Together, these strategies reduce false positives and leverage self-supervised learning to ensure effective detection with minimal labeled data, setting TCG-IDS apart from previous GNN-based IDS solutions.

The main contributions are summarized as follows:

- TCG-IDS is the first IDS to integrate temporal graph learning with self-supervised contrastive learning, leveraging temporal dynamics and spatial dependencies to enhance detection robustness.
- TCG-IDS employs three novel contrastive learning strategies, temporal, asymmetric, and masked contrasting, to enhance robustness. These enable the model to capture temporal network dynamics, node relationships, and learn from incomplete data, ensuring effective detection with limited labeled data.
- TCG-IDS models traffic as a continuous dynamic graph with temporal encoding, enabling real-time, fine-grained detection. This captures evolving network behaviors, ensuring accurate and timely intrusion detection.
- Comprehensive experiments on TCG-IDS are performed on two widely-used datasets to evaluate its performance. Results demonstrate that TCG-IDS significantly outperforms state-of-the-art graph learning models and IDSs, achieving a balanced accuracy of 99.48% and 91.48% for binary detection on the two datasets, respectively.

II. Overview

A. Threat Model

This paper focuses on an enterprise network environment where attackers, either external (e.g., malicious hackers) or internal (e.g., disgruntled employees), attempt to compromise the confidentiality, integrity, or availability of network resources. These attackers exploit vulnerabilities in the network infrastructure, such as misconfigured firewalls, unpatched systems, or weak authentication mechanisms, to gain unauthorized access, as illustrated in Figure 1. Once inside, they may perform malicious activities like scanning for vulnerable hosts, escalating privileges, exfiltrating sensitive data, or launching denial-of-service attacks. To counter these threats, the paper assumes the deployment of a network intrusion detection system (IDS) that monitors network traffic for signs of malicious activity. IDS, positioned at strategic network points, utilizes TCG-IDS, a framework that employs temporal contrastive learning and graph neural networks to learn robust representations of network entities and interactions over time. These learned representations are then used by the detector

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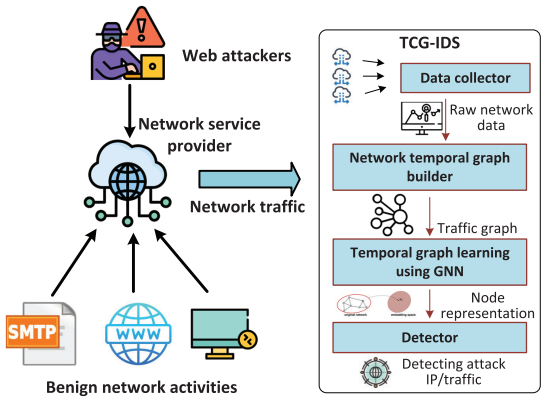


Fig. 1. Threat model.

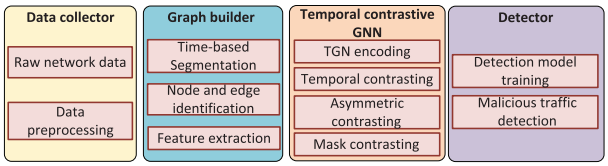


Fig. 2. Overview of TCG-IDS.

to identify potential intrusions, with the IDS continuously updated based on feedback from security analysts and evolving threat intelligence.

B. System Overview

As depicted in Figure 2, TCG-IDS consists of four main components: data collector, graph builder, temporal contrastive GNN, and detector. Data collector is responsible for collecting and preprocessing raw network traffic data from various sources, such as packet captures or flow records, and extracting relevant features for each network entity, such as IP addresses or port numbers. Graph builder then constructs a sequence of attributed temporal dynamic graphs from the preprocessed data, where each graph snapshot represents the state of the network at a specific time step, with nodes representing network entities and edges representing their interactions. Temporal contrastive GNN is the core component of TCG-IDS, which learns robust representations of the network entities by capturing their spatial and temporal dependencies using a combination of graph neural networks, temporal contrastive learning, asymmetric contrastive learning, and masked contrastive learning. Finally, detector utilizes the learned representations to identify malicious network behavior, such as malicious activities or intrusions, by employing supervised detection classifier.

III. Design of TCG-IDS

A. Data Collector

TCG-IDS captures and preprocesses network traffic data from various sources, e.g., routers, switches, and monitoring tools, to create a structured dataset for intrusion detection. Using flow records like NetFlow, sFlow, or IPFIX, it extracts key attributes, including IP addresses, port numbers, protocols, transferred bytes, packets, duration, and timestamps. Advanced features, e.g., TCP flags, Type of Service (ToS), Autonomous System (AS) numbers, and application layer information, are also included when available. Preprocessing ensures data quality through cleaning, normalization, encoding, and sampling, addressing incomplete or duplicate records and scaling feature values. The resulting dataset, with rows representing network flows and columns as features, is prepared for constructing the temporal dynamic graphs used by TCG-IDS.

B. Graph Builder

The framework constructs a sequence of temporal attributed graphs from the preprocessed network flow data, capturing both spatial and temporal dependencies to facilitate robust and discriminative node representations for effective intrusion detection. The data is divided into T discrete time steps, each corresponding to a specific time window Δt, selected based on the distribution of timestamps and empirical validation to balance temporal resolution and computational feasibility. At each time step t, unique IP addresses are identified to form the node set V_t of the graph snapshot G_t. Each node $v_i^t \in V_t$ represents a device in the network at time t, with relevant features extracted into a feature vector $x_i^t \in \mathbb{R}^d$, where d denotes the number of node features.

Edges E_t between nodes in V_t are identified based on network flows, with an edge $e_{ij}^t \in E_t$ existing if there is at least one flow between the corresponding IP addresses within the time window t. Relevant features for each edge e_{ij}^t are extracted into a feature vector $y_{ij}^t \in \mathbb{R}^m$, where m is the number of edge features. Common edge features include total bytes and packets transferred, duration, average bytes per packet, average packets per second, ratio of incoming to outgoing bytes/packets, protocol distribution, and flag distribution. A graph snapshot $G_t = (V_t, E_t)$ is constructed for each time step t, associating node features x_i^t and edge features y_{ij}^t with their respective nodes and edges. The sequence of T graph snapshots $G = \{G_1, G_2, \ldots, G_T\}$, representing the temporal evolution of the network traffic, is outputted. Each graph snapshot G_t is represented as (V_t, E_t, X_t, Y_t), where $X_t \in \mathbb{R}^{|V_t| \times d}$ is the matrix of node features and $Y_t \in \mathbb{R}^{|E_t| \times m}$ is the matrix of edge features. This sequence of temporal attributed graphs is then used for node representation learning and detection, effectively capturing the spatial and temporal dependencies in the network data. This sequence of temporal attributed graphs is then used for node representation learning and detection, effectively capturing the spatial and temporal dependencies in the network data.

C. Temporal Contrastive GNN

This module learns robust node representations from temporal attributed graphs, capturing spatial and temporal dependencies for effective detection through GNN encoding, temporal contrasting, asymmetric contrasting, and masked contrasting.

1) GNN Encoding: The GNN Encoder is responsible for learning node embeddings for each temporal graph snapshot G_t. Given the dynamic nature of the network, TCG-IDS

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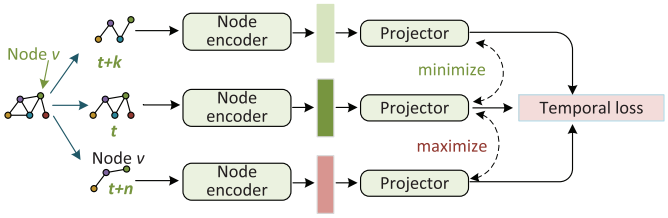


Fig. 3. Illustration of temporal contrasting.

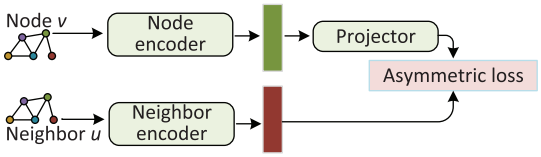


Fig. 4. Illustration of asymmetric contrasting.

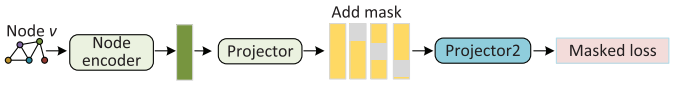


Fig. 5. Illustration of masked contrasting.

utilizes a Temporal Graph Network (TGN) encoder [21] to capture the temporal evolution of node features and interactions effectively. It combines memory modules and graph-based operators to process continuous-time dynamic graphs. For a given graph snapshot $G_t = (V_t, E_t, X_t)$, the TGN encoder computes the node embeddings h_i^t for each node $v_i^t \in V_t$ as follows:

$$h_i^t = \text{TGN}_\theta(G_t, X_t)[i] \in \mathbb{R}^D \tag{1}$$

where TGN_θ is the TGN encoder with learnable parameters θ, and D is the dimensionality of the node embeddings. The TGN encoder leverages a combination of memory modules and graph attention mechanisms to capture both the historical context and the immediate neighborhood information of each node. The memory of each node is updated upon every interaction, allowing the TGN encoder to maintain long-term dependencies and provide up-to-date embeddings even in the presence of infrequent interactions. This ensures that the embeddings reflect both the temporal and structural characteristics of the network.

2) Temporal Contrasting: This step aims to learn temporally consistent node representations by contrasting the embeddings of a node at different time steps. The goal is to maximize the similarity between the embeddings of a node at nearby time steps while minimizing the similarity between the embeddings of a node at distant time steps. As illustrated in Figure 3, for each node v_i, it samples a positive pair (h_i^t, h_i^{t+k}) and a negative pair (h_i^t, h_i^{t+n}), where k is a small time lag and n is a larger time lag, ensuring that the time steps are within the valid range and the node v_i has varied significantly. The temporal contrastive loss $\mathcal{L}_{\text{temp}}$ is defined as Eq. 2.

$$\mathcal{L}_{\text{temp}} = \frac{1}{|V|} \sum_{v_i \in V} \sum_{t=1}^{T-\max(k,n)} \max(0, \Delta + \text{sim}(h_i^t, h_i^{t+n}) - \text{sim}(h_i^t, h_i^{t+k})) \tag{2}$$

where Δ is the margin, $\text{sim}(\cdot,\cdot)$ is a cosine similarity function, and $|V|$ is the total number of nodes across all time steps.

3) Asymmetric Contrasting: It aims to learn node representations that capture the local neighborhood context and the asymmetric relationships between nodes. It predicts the embeddings of neighboring nodes using an asymmetric predictor function. As illustrated in Figure 4, for each node v_i^t at time step t, it samples a set of neighboring nodes $\mathcal{N}(v_i^t)$ based on the graph structure. The asymmetric predictor function f_ϕ takes the embedding of the node v_i^t as input and predicts the embeddings of its neighboring nodes:

$$\hat{h}_j^t = f_\phi(h_i^t) \in \mathbb{R}^D, \quad \forall v_j^t \in \mathcal{N}(v_i^t) \tag{3}$$

The asymmetric contrastive loss $\mathcal{L}_{\text{asym}}$ is formulated as:

$$\mathcal{L}_{\text{asym}} = \frac{1}{|V|} \sum_{t=1}^{T} \sum_{v_i^t \in V_t} \sum_{v_j^t \in \mathcal{N}(v_i^t)} \|\hat{h}_j^t - h_j^t\|_2^2 \tag{4}$$

where $\|\cdot\|_2$ denotes the L2 norm.

4) Masked Contrasting: To learn node representations that capture multi-facet features of the network traffic, it divides the node embeddings into multiple factors, masks each factor separately, and performs contrastive learning on the masked embeddings. As illustrated in Figure 5, for each node v_i^t at time step t, it divides its embedding h_i^t into M factors:

$$h_i^t = [c_{i1}^t, c_{i2}^t, \ldots, c_{iM}^t] \tag{5}$$

where $c_{im}^t \in \mathbb{R}^{D/M}$ is the m-th factor of the embedding. It generates M masked embeddings by replacing each factor with a learnable mask token `[MASK]`, formulated as Eq. 6:

$$\tilde{h}_{im}^t = [c_{i1}^t, \ldots, \texttt{[MASK]}, \ldots, c_{iM}^t], \quad m = 1, \ldots, M \tag{6}$$

For optimizing the model, the masked contrastive loss $\mathcal{L}_{\text{mask}}$ is defined as:

$$\mathcal{L}_{\text{mask}} = \frac{1}{|V|} \sum_{t=1}^{T} \sum_{v_i^t \in V_t} \sum_{m=1}^{M} \max(0, \Delta + \text{sim}(\tilde{h}_{im}^t, \tilde{h}_{jm}^t) - \text{sim}(\tilde{h}_{im}^t, h_i^t)) \tag{7}$$

where v_j^t is a randomly sampled node from the same time step t, and $\tilde{h}_{jm}^t$ is its corresponding masked embedding.

5) Training Objective: The overall training objective of the Temporal Contrastive GNN-based Learner is a combination of the temporal contrastive loss, the asymmetric contrastive loss, and the masked contrastive loss: $\mathcal{L} = \mathcal{L}_{\text{temp}} + \lambda_1 \mathcal{L}_{\text{asym}} + \lambda_2 \mathcal{L}_{\text{mask}}$, where λ_1 and λ_2 are hyperparameters that balance the contribution of the different loss terms. The Temporal Contrastive GNN-based Learner is trained end-to-end using backpropagation and stochastic gradient descent optimization. The learned node embeddings h_i^t are then used as input to the next module for identifying anomalous network behavior. By incorporating temporal contrasting, asymmetric contrasting, and masked contrasting, the Temporal Contrastive GNN-based

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Algorithm 1 Training Process of TCG-IDS

Input: Sequence of graph snapshots $G = \{G_1, G_2, \ldots, G_T\}$, learning rate η, epochs E
Output: Trained model parameters θ
for epoch = 1 to E **do**
  Initialize gradient accumulators $d\theta \leftarrow 0$
  for t = 1 to T **do**
    Extract $G_t = (V_t, E_t, X_t, Y_t)$
    for each node $v_i^t \in V_t$ **do**
      $h_i^t \leftarrow \text{TGN}_\theta(G_t, X_t)[i]$
  for each node v_i **do**
    for t = 1 to $T - \max(k, n)$ **do**
      Sample pairs (h_i^t, h_i^{t+k}) and (h_i^t, h_i^{t+n})
      Compute temporal contrastive loss $\mathcal{L}_{\text{temp}}$
  for each node $v_i^t \in V_t$ **do**
    for each neighboring node $v_j^t \in \mathcal{N}(v_i^t)$ **do**
      Predict neighbor embeddings $\hat{h}_j^t \leftarrow f_\phi(h_i^t)$
      Compute asymmetric contrastive loss $\mathcal{L}_{\text{asym}}$
  for each node $v_i^t \in V_t$ **do**
    for m = 1 to M **do**
      $\tilde{h}_{im}^t \leftarrow [c_{i1}^t, \ldots, \texttt{[MASK]}, \ldots, c_{iM}^t]$
      $\tilde{h}_{jm}^t \leftarrow [c_{j1}^t, \ldots, \texttt{[MASK]}, \ldots, c_{jM}^t]$
      Compute masked contrastive loss $\mathcal{L}_{\text{mask}}$
  $\mathcal{L} \leftarrow \mathcal{L}_{\text{temp}} + \lambda_1 \mathcal{L}_{\text{asym}} + \lambda_2 \mathcal{L}_{\text{mask}}$
  Update parameters $\theta \leftarrow \theta - \eta \cdot \nabla_\theta \mathcal{L}$

Learner is able to learn node representations that capture the spatial, temporal, and multi-facet dependencies in the network traffic data, enabling more effective detection in TCG-IDS. Algorithm 1 details this training process.

D. Detector

The detection module identifies malicious network behavior based on the node embeddings $\{h_i^t\}$ learned from the Temporal Contrastive GNN-based Learner. Each embedding $h_i^t \in \mathbb{R}^D$ represents node v_i at time t. The module outputs a score $s_i^t \in [0, 1]$, where a higher score indicates a higher likelihood of malicious activity. This study employs a supervised approach using a LightGBM classifier, which was chosen for its efficiency in handling large-scale datasets, its ability to model complex, non-linear relationships in the data, and its faster training time compared to other gradient boosting methods. While this framework supports unsupervised methods such as One-Class SVM and Isolation Forest for scenarios with limited labeled data, this work emphasizes the supervised setup, with future directions focusing on semi-supervised methods to effectively utilize unlabeled data and minimal labeled samples.

IV. Performance Evaluation

A. Experimental Setting

1) Dataset: This study evaluates TCG-IDS using two publicly available network traffic datasets: NF-CSE-CIC-IDS2018-V2 [22] and NF-UNSW-NB15-v2 [23]. NF-CSE-CIC-IDS2018-V2, developed by the Canadian Institute for Cybersecurity, contains 18,893,623 records collected over 10 days in July 2018, including 10,423,751 attack samples covering DDoS, DoS, Botnet, Brute Force, Infiltration, and Web attacks, alongside 8,469,872 normal traffic samples. It provides rich features, e.g., statistical metrics and connection details for advanced intrusion detection model evaluation. NF-UNSW-NB15-v2, created by the Australian Centre for Cyber Security, includes 2,391,775 records collected over two months in early 2015, with 96,553 attack samples spanning categories like Exploits, Fuzzers, Reconnaissance, and DoS, and 2,295,222 normal traffic samples. It offers detailed features extracted from raw network packets, including payloads and flow statistics, enabling robust evaluation of intrusion detection models.

To address the absence of explicit timestamps in the datasets, we generated timestamps using `FLOW_DURATION_MILLISECONDS` feature. The process began by assigning the first record a timestamp of 0 milliseconds as a reference point. Records were then sorted based on their occurrence in the original data file to maintain a logical temporal sequence. For each subsequent record i, the timestamp T_i was calculated as the cumulative sum of the `FLOW_DURATION_MILLISECONDS` values of all preceding records, i.e., $T_i = T_{i-1} + \text{FLOW_DURATION_MILLISECONDS}_{i-1}$, with $T_0 = 0$. For records with missing or zero `FLOW_DURATION_MILLISECONDS` values, the same timestamp as the preceding record was assigned to preserve continuity. This approach ensures a realistic temporal context, enabling a chronological train-test split that respects the temporal dynamics of network traffic.

2) Parameters: The encoder uses a 128-dimensional embedding, 256-dimensional memory, and two graph attention layers with 8 heads each, employing ReLU and a 0.5 dropout rate. Temporal dynamics are captured through time embeddings. The projector network comprises two 128-unit fully connected layers with ReLU and batch normalization. Temporal contrasting uses a margin (Δ) of 1.0 with time lags $k = 5$ and $n = 10$, while asymmetric contrasting involves a predictor with two 128-unit layers. Masked contrasting divides node embeddings into four factors, masking one at a time. The training combines temporal, asymmetric, and masked contrastive losses ($\lambda_1, \lambda_2 = 0.5$) and uses the Adam optimizer ($lr = 0.001$, batch size 64) with gradient clipping and early stopping to ensure effective detection in dynamic networks.

3) Baseline: To benchmark TCG-IDS, we compare it against state-of-the-art graph-based models, including GraphSAGE [16], GAT [15], and TGN [21], which are widely used for network intrusion detection and related tasks. These models were selected for their graph-focused methodologies, complementing TCG-IDS's temporal contrastive approach. GraphSAGE aggregates features from a node's local neighborhood, GAT employs attention mechanisms to weigh neighbors' importance, and TGN captures temporal evolution in dynamic graphs. We utilized the same hyperparameters as TCG-IDS, tuned through grid search for optimal performance, to ensure a fair comparison. We focuses on comparing GNN-based techniques, distinct from non-GNN-based approaches. All evaluations were conducted on an Intel Xeon Platinum 8336C CPU and an NVIDIA GeForce RTX 4090 GPU, with the running time per batch recorded to assess computational efficiency for large-scale network data.

The evaluation uses a chronological split, with 70% of earlier timestamps for training and 30% later for testing, preserving temporal integrity. Models are trained on graph snapshots, with metrics including precision, recall, F1-score, balanced accuracy (BAC), true negative rate (TNR), and true

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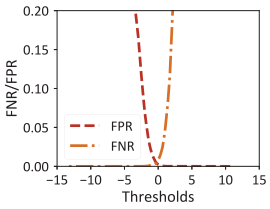


Fig. 6. FPR/FNR under NF-CSE-CIC-IDS2018-v2.

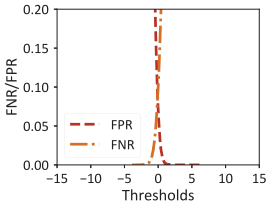


Fig. 7. FPR/FNR under NF-UNSW-NB15-v2.

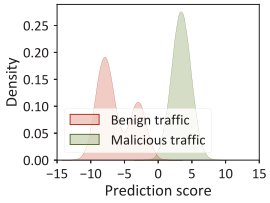


Fig. 8. Distribution of prediction score under NF-CSE-CIC-IDS2018-v2.

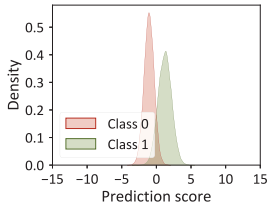


Fig. 9. Distribution of prediction score under NF-UNSW-NB15-v2.

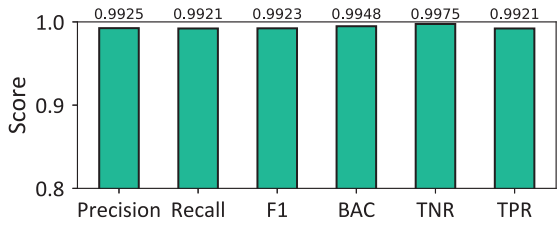


Fig. 10. Overall performance under NF-CSE-CIC-IDS2018-v2.

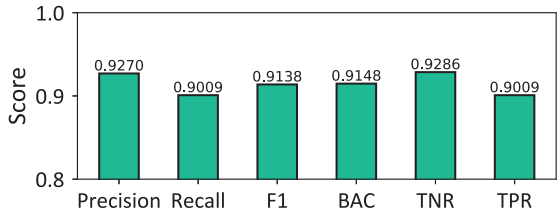


Fig. 11. Overall performance under NF-UNSW-NB15-v2.

positive rate (TPR) for binary detection, and mean false positive rate (mFPR) for multi-class detection.

B. Overall Performance

1) Binary Classification: The NF-CSE-CIC-IDS2018-V2 dataset was used to evaluate the effectiveness of TCG-IDS in identifying diverse attack scenarios. This dataset includes a wide variety of attack types, such as DDoS, DoS, Botnet, Brute Force, Infiltration, and Web attacks, making it a comprehensive benchmark for intrusion detection systems. As shown in Figure 10, TCG-IDS achieved outstanding performance on this dataset, with a precision of 99.25%, recall of 99.21%, F1-score of 99.23%, BAC of 99.48%, TNR of 99.75%, and TPR of 99.21%. These results highlight the system's capability to accurately distinguish between normal and malicious traffic, ensuring high detection rates with minimal false positives. The high precision and recall values indicate that TCG-IDS is highly effective in identifying attacks without raising many false alarms, which is crucial for real-world deployment. Figure 6 presents FPR and FNR varied with different thresholds under the NF-CSE-CIC-IDS2018-V2 dataset in detail. The FNR is 0.068% when the FPR is 3%. Figure 8 also presents the prediction score distribution of TCG-IDS under the NF-CSE-CIC-IDS2018-V2 dataset in detail.

Similarly, the NF-UNSW-NB15-v2 dataset, which includes a mix of normal traffic and various attack types such as Exploits, Fuzzers, Reconnaissance, DoS, Analysis, Backdoor, Shellcode, and Worms, was used to further validate the robustness of TCG-IDS. As shown in Figure 11, on this dataset, TCG-IDS achieved a precision of 92.70%, recall of 90.09%, F1-score of 91.38%, BAC of 91.48%, TNR of 92.86%, and TPR of 90.09%. Although the performance is slightly lower than on the NF-CSE-CIC-IDS2018-V2 dataset, these results still demonstrate robust detection capabilities across different network environments and attack scenarios. The balanced accuracy and recall values indicate that the system can effectively identify malicious activities while maintaining a low false negative rate, ensuring comprehensive network protection. Figure 7 presents FPR and FNR varied with different thresholds under the NF-UNSW-NB15-v2 dataset in detail. The FNR is 16.71% when the FPR is 3%. Figure 9 also presents the prediction score distribution of TCG-IDS under the NF-UNSW-NB15-v2 dataset in detail.

2) Multi-Class Classification: In addition to binary classification, the TCG-IDS system's performance is evaluated in multi-class detection scenarios using the NF-CSE-CIC-IDS2018-V2 and NF-UNSW-NB15-v2 datasets. The evaluation settings involved training the TCG-IDS system on a subset of each dataset and testing its performance on the remaining data, ensuring the results reflect the system's real-world capability to handle diverse and complex attack scenarios. The metrics used for this evaluation include a FPR for each class.

For the NF-CSE-CIC-IDS2018-V2 dataset, the TCG-IDS system was evaluated on ten different classes, including Benign, DDOS attack-HOIC, DoS attacks-Hulk, DDoS attacks-LOIC-HTTP, Bot, Infiltration, SSH-Bruteforce,

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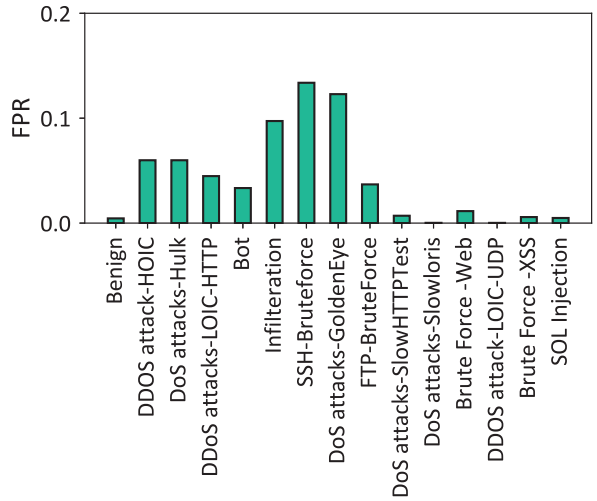


Fig. 12. Multi-class FPR under NF-CSE-CIC-IDS2018-v2.

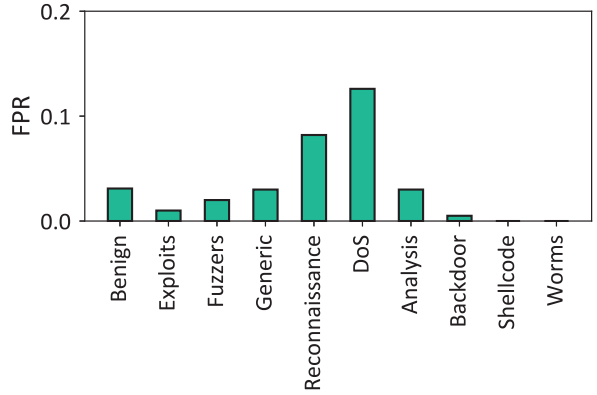


Fig. 13. Multi-class FPR under NF-UNSW-NB15-v2.

DoS attacks-GoldenEye, FTP-BruteForce, DoS attacks-SlowHTTPTest, DoS attacks-Slowloris, Brute Force-Web, DDOS attack-LOIC-UDP, Brute Force-XSS, and SQL Injection. The system achieved the following FPR for each class, as presented in Figure 12. The results highlight TCG-IDS's proficiency in correctly identifying various attack types while maintaining a low FPR. The system shows particularly low FPRs for critical classes such as Benign (0.45%), DoS attacks-Slowloris (0.02%), DDOS attack-LOIC-UDP (0.02%), and SQL Injection (0.49%), which are crucial for reducing false alarms and ensuring accurate threat detection. However, higher FPRs for classes like SSH-Bruteforce (13.37%) and DoS attacks-GoldenEye (12.29%) indicate potential areas for improvement. These higher FPRs might be due to the complexity and similarity of these attacks to normal traffic, requiring more sophisticated feature extraction or advanced modeling techniques to enhance detection accuracy.

For the NF-UNSW-NB15-v2 dataset, TCG-IDS was evaluated on fifteen different classes, including Benign, Exploits, Fuzzers, Generic, Reconnaissance, DoS, Analysis, Backdoor, Shellcode, and Worms. The FPR for each class is presented

TABLE I

NUMBER OF MULTI-CLASS SAMPLES AND ITS PERCENTAGE OF NF-CSE-CIC-IDS2018-v2

Types	# Samples	Percentage
Benign	16,635,567	88.05%
DDoS attack-HOIC	1,080,858	5.72%
DoS attacks-Hulk	432,648	2.29%
DDoS attacks-LOIC-HTTP	307,300	1.63%
Bot	143,097	0.76%
Infiltration	116,361	0.62%
SSH-Bruteforce	94,979	0.50%
DoS attacks-GoldenEye	27,723	0.15%
FTP-BruteForce	25,933	0.14%
DoS attacks-SlowHTTPTest	14,116	0.07%
DoS attacks-Slowloris	9,512	0.05%
Brute Force -Web	2,143	0.01%
DDoS attack-LOIC-UDP	2,112	0.01%
Brute Force -XSS	927	0.00%
SQL Injection	432	0.00%

TABLE II

NUMBER OF MULTI-CLASS SAMPLES AND ITS PERCENTAGE OF NF-UNSW-NB15-v2 DATASET

Types	# Samples	Percentage
Benign	2,295,222	96.02%
Exploits	31,551	1.32%
Fuzzers	22,310	0.93%
Generic	16,560	0.69%
Reconnaissance	12,779	0.53%
DoS	5,794	0.24%
Analysis	2,299	0.10%
Backdoor	2,169	0.09%
Shellcode	1,427	0.06%
Worms	164	0.01%

in Figure 13. The results indicate that TCG-IDS maintains a low FPR across most classes, with particularly low rates for critical classes such as Exploits (1.07%), Fuzzers (2.08%), and Backdoor (0.51%), demonstrating its ability to accurately classify different types of network attacks. Slightly higher FPR for classes such as DoS (12.61%) and Reconnaissance (8.20%) suggests areas for further refinement but still indicates strong overall performance. These higher FPRs could be due to the more varied and complex nature of these attacks, which might require additional feature engineering or model tuning to improve detection accuracy.

Overall, the multi-class detection results for both datasets demonstrate that TCG-IDS can effectively distinguish between various attack types, maintaining low FPRs across different classes. This capability is essential for providing comprehensive network security, as it allows for the precise identification and mitigation of specific threats. The system's ability to maintain high accuracy and low FPRs across multiple classes underscores its robustness and effectiveness in real-world network intrusion detection scenarios. These findings suggest that TCG-IDS is a reliable and effective tool for enhancing network security through precise and accurate multi-class detection.

C. Ablation Study

To understand the contributions of different components in TCG-IDS, an ablation study is conducted by systematically removing each component and evaluating the performance of the modified system. The components evaluated include temporal contrasting, asymmetric contrasting, and masked contrasting. The results of the ablation study on the

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TABLE III
BINARY CLASSIFICATION PERFORMANCE OF ABLATION STUDY UNDER NF-CSE-CIC-IDS2018-V2 DATASET

Method	Precision	Recall	F1	BAC	TPR	TNR
TCG-IDS w/o temporal contrasting	88.74%	85.44%	87.06%	87.27%	89.10%	85.44%
TCG-IDS w/o asymmetric Contrasting	90.05%	87.12%	88.56%	88.72%	90.32%	87.12%
TCG-IDS w/o masked Contrasting	95.28%	93.45%	94.35%	94.40%	95.34%	93.45%
TCG-IDS	99.25%	99.21%	99.23%	99.48%	99.75%	99.21%

TABLE IV
BINARY CLASSIFICATION PERFORMANCE OF ABLATION STUDY UNDER NF-UNSW-NB15-V2 DATASET

Method	Precision	Recall	F1	BAC	TPR	TNR
TCG-IDS w/o temporal contrasting	82.92%	80.33%	81.60%	81.83%	83.34%	80.33%
TCG-IDS w/o asymmetric Contrasting	85.77%	83.05%	84.39%	84.59%	86.13%	83.05%
TCG-IDS w/o masked Contrasting	88.16%	85.90%	87.02%	87.14%	88.39%	85.90%
TCG-IDS	92.70%	90.09%	91.38%	91.48%	92.86%	90.09%

TABLE V
MULTI-CLASS CLASSIFICATION MFPRS OF ABLATION STUDY UNDER NF-CSE-CIC-IDS2018-V2

Method	mFPR
TCG-IDS w/o temporal contrasting	19.29%
TCG-IDS w/o asymmetric Contrasting	14.26%
TCG-IDS w/o masked Contrasting	8.58%
TCG-IDS	3.34%

TABLE VI
MULTI-CLASS CLASSIFICATION MFPRS OF ABLATION STUDY UNDER NF-UNSW-NB15-V2

Method	mFPR
TCG-IDS w/o temporal contrasting	25.36%
TCG-IDS w/o asymmetric Contrasting	13.18%
TCG-IDS w/o masked Contrasting	5.19%
TCG-IDS	4.15%

NF-CSE-CIC-IDS2018-V2 dataset are presented in Table III. Removing temporal contrasting resulted in a significant drop in performance, with precision, recall, and F1-score decreasing to 88.74%, 85.44%, and 87.06% respectively. Similarly, excluding asymmetric contrasts led to a reduction in precision, recall, and F1-score to 90.05%, 87.12%, and 88.56%. When masked contrasting was removed, the system's performance also declined, though it remained relatively high with a precision of 95.28%, recall of 93.45%, and F1-score of 94.35%. The full TCG-IDS system, which includes all components, achieved the highest performance with precision, recall, and F1-score of 99.25%, 99.21%, and 99.23% respectively, indicating the importance of each component in enhancing the overall effectiveness of the system.

The ablation study results on the NF-UNSW-NB15-v2 dataset, shown in Table IV, exhibit a similar trend. Excluding temporal contrasting resulted in the precision, recall, and F1-score dropping to 82.92%, 80.33%, and 81.60% respectively. Removing asymmetric contrasts led to performance decreases with precision, recall, and F1-score at 85.77%, 83.05%, and 84.39%. Without masked contrasting, the system's precision, recall, and F1-score were 88.16%, 85.90%, and 87.02%. The complete TCG-IDS system outperformed these modified versions, achieving precision, recall, and F1-score of 92.70%, 90.09%, and 91.38%, respectively. These results reinforce the critical role each component plays in maintaining high detection performance across different datasets.

For multi-class detection, the mean false positive rates (mFPRs) for each ablation variant on the NF-CSE-CIC-IDS2018-V2 and NF-UNSW-NB15-v2 datasets are provided in Tables V and VI. On the NF-CSE-CIC-IDS2018-V2 dataset, the mFPR increased significantly when temporal contrasting was removed (19.29%), followed by asymmetric contrasting (14.26%), and masked contrasting (8.58%). The complete TCG-IDS system achieved the lowest mFPR (3.34%), underscoring the importance of all components in maintaining a low rate of false positives across multiple classes. Similarly, on the NF-UNSW-NB15-v2 dataset, removing temporal contrasting led to the highest mFPR (25.36%), followed by removing asymmetric contrasting (13.18%), and masked contrasting (5.49%). The full system maintained a low mFPR (4.15%), indicating robust performance in detecting various attack types.

TABLE VII
COMPARING WITH BASELINE MODELS UNDER NF-CSE-CIC-IDS2018-V2 DATASET IN BINARY DETECTION TASK

Method	Precision	Recall	F1	BAC	TPR	TNR
GraphSage	72.91%	68.51%	70.64%	71.46%	74.41%	68.51%
GAT	79.10%	74.35%	76.65%	77.30%	80.25%	74.35%
TGN	84.14%	79.97%	82.01%	82.41%	84.85%	79.97%
TCG-IDS (Ours)	99.25%	99.21%	99.23%	99.48%	99.75%	99.21%

D. Comparing With Baseline Models

To evaluate the effectiveness of TCG-IDS over existing models, a comparison with several baseline models is performed, including GraphSage, GAT, and TGN, on the NF-CSE-CIC-IDS2018-V2 and NF-UNSW-NB15-v2 datasets. The comparison focused on binary detection tasks, using metrics such as precision, recall, F1-score, BAC, TPR, and TNR. As shown in Table VII, TCG-IDS significantly outperformed the baseline models on the NF-CSE-CIC-IDS2018-V2 dataset. GraphSage achieved a precision of 72.91%, recall of 68.51%, and F1-score of 70.64%, while GAT improved these metrics to 79.10%, 74.35%, and 76.65%, respectively. TGN further enhanced performance with a precision of 84.14%, recall of 79.97%, and F1-score of 82.01%. However, TCG-IDS achieved the highest precision of 99.25%, recall of 99.21%,

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TABLE VIII
COMPARING WITH BASELINE MODELS UNDER NF-UNSW-NB15-V2 DATASET IN BINARY DETECTION TASK

Method	Precision	Recall	F1	BAC	TPR	TNR
GraphSage	70.17%	66.69%	68.39%	69.10%	71.50%	66.69%
GAT	74.41%	70.03%	72.15%	72.91%	75.79%	70.03%
TGN	81.67%	77.36%	79.46%	79.96%	82.55%	77.36%
TCG-IDS (Ours)	92.70%	90.09%	91.38%	91.48%	92.86%	90.09%

TABLE IX
COMPARING WITH BASELINE MODELS UNDER NF-CSE-CIC-IDS2018-V2 IN MULTI-CLASS DETECTION TASK

Method	mFPR
GraphSage	28.70%
GAT	16.82%
TGN	13.21%
TCG-IDS (Ours)	3.34%

TABLE X
COMPARING WITH BASELINE MODELS UNDER NF-UNSW-NB15-V2 IN MULTI-CLASS DETECTION TASK

Method	mFPR
GraphSage	31.48%
GAT	26.54%
TGN	16.92%
TCG-IDS (Ours)	4.15%

TABLE XI
COMPARING WITH EXISTING METHODS UNDER NF-CSE-CIC-IDS2018-V2 DATASET IN BINARY DETECTION TASK

Method	Precision	Recall	F1	BAC	TPR	TNR
DLGNN [18]	83.96%	80.36%	82.12%	82.47%	84.57%	80.36%
EgraphSage [19]	93.27%	91.21%	92.23%	92.30%	93.38%	91.21%
Sarhan et al. [7]	89.11%	77.58%	82.95%	84.03%	90.48%	77.58%
TCG-IDS (Ours)	99.25%	99.21%	99.23%	99.48%	99.75%	99.21%

TABLE XII
COMPARING WITH EXISTING METHODS UNDER NF-UNSW-NB15-V2 DATASET IN BINARY DETECTION TASK

Method	Precision	Recall	F1	BAC	TPR	TNR
DLGNN [18]	79.69%	75.19%	77.38%	77.97%	80.75%	75.19%
EgraphSage [19]	90.90%	87.96%	89.40%	89.55%	91.15%	87.96%
Sarhan et al. [7]	92.23%	78.05%	84.55%	85.72%	93.4%	78.05%
TCG-IDS (Ours)	92.70%	90.09%	91.38%	91.48%	92.86%	90.09%

and F1-score of 99.23%, demonstrating superior detection capabilities and robustness.

In the binary detection task, TCG-IDS demonstrated superior performance on both the NF-CSE-CIC-IDS2018-V2 and NF-UNSW-NB15-v2 datasets. On NF-CSE-CIC-IDS2018-V2, TCG-IDS achieved a precision of 99.25%, recall of 99.21%, and F1-score of 99.23%, significantly outperforming DLGNN, EgraphSage, and Sarhan et al., as shown in Table XI. Similarly, on NF-UNSW-NB15-v2, TCG-IDS attained a precision of 92.70%, recall of 90.09%, and F1-score of 91.38%, surpassing these baseline models (Table XII). In the multi-class detection task, TCG-IDS achieved the lowest mean false positive rates (mFPRs) of 3.34% on NF-CSE-CIC-IDS2018-V2 (Table XIII) and 4.15% on NF-UNSW-NB15-v2 (Table XIV), compared to higher mFPRs for DLGNN, EgraphSage, and Sarhan et al. These results highlight TCG-IDS's robustness, precision, and ability to balance high detection accuracy with low false positive rates, outperforming existing methods across diverse network intrusion scenarios.

TABLE XIII
COMPARING WITH EXISTING METHODS UNDER NF-CSE-CIC-IDS2018-V2 IN MULTI-CLASS DETECTION TASK

Method	mFPR
DLGNN [18]	13.21%
EgraphSage [19]	5.62%
Sarhan et al. [7]	8.13%
TCG-IDS (Ours)	3.34%

TABLE XIV
COMPARING WITH EXISTING METHODS UNDER NF-UNSW-NB15-V2 IN MULTI-CLASS DETECTION TASK

Method	mFPR
DLGNN [18]	20.70%
EgraphSage [19]	13.25%
Sarhan et al. [7]	7.17%
TCG-IDS (Ours)	4.15%

TABLE XV
RUNNING TIME (SECOND) OF TCG-IDS PER BATCH

Dataset	Running time
NF-CSE-CIC-IDS2018-v2	0.3754
NF-UNSW-NB15-v2	0.5112

E. Comparing With Existing Methods

To evaluate the performance of TCG-IDS in comparison to existing methods, experiments are conducted using the NF-CSE-CIC-IDS2018-V2 and NF-UNSW-NB15-v2 datasets in both binary and multi-class detection tasks. The existing methods compared include DLGNN [18], EgraphSage [19], and Sarhan et al. [7], which are representative of recent advances in graph-based intrusion detection systems.

In the binary detection task, TCG-IDS demonstrated superior performance on both the NF-CSE-CIC-IDS2018-V2 and NF-UNSW-NB15-v2 datasets. On NF-CSE-CIC-IDS2018-V2, TCG-IDS achieved a precision of 99.25%, recall of 99.21%, and F1-score of 99.23%, significantly outperforming DLGNN, EgraphSage, and Sarhan et al., as shown in Table XI. Similarly, on NF-UNSW-NB15-v2, TCG-IDS attained a precision of 92.70%, recall of 90.09%, and F1-score of 91.38%, surpassing these baseline models (Table XII). In the multi-class detection task, TCG-IDS achieved the lowest mean false positive rates (mFPRs) of 3.34% on NF-CSE-CIC-IDS2018-V2 (Table XIII) and 4.15% on NF-UNSW-NB15-v2 (Table XIV), compared to higher mFPRs for DLGNN, EgraphSage, and Sarhan et al. These results highlight TCG-IDS's robustness, precision, and ability to balance high detection accuracy with low false positive rates, outperforming existing methods across diverse network intrusion scenarios.

F. Running Time Evaluation

The running time of TCG-IDS per batch is measured to assess its computational efficiency. The experiments were conducted on two publicly available network traffic datasets, NF-CSE-CIC-IDS2018-V2 and NF-UNSW-NB15-V2. Each batch consisted of graph snapshots processed by the TCG-IDS

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TABLE XVI
TIME COMPLEXITY OF TCG-IDS

Component	Complexity
TGN Encoder	$O(N_v + N_e)$
Temporal Contrasting	$O(N_v \cdot T \cdot D)$
Asymmetric Contrasting	$O(N_v \cdot \bar{d} \cdot D)$
Masked Contrasting	$O(N_v \cdot M \cdot D)$
Overall Complexity	$O(T \cdot (N_v + N_e)) + O(N_v \cdot T \cdot D) + O(N_v \cdot \bar{d} \cdot D) + O(N_v \cdot M \cdot D)$
Simplified Complexity	$O(N_v \cdot (T + \bar{d} \cdot D + M \cdot D) + N_e)$

system. Results of the running time evaluation are presented in Table XV. For NF-CSE-CIC-IDS2018-V2 dataset, it achieved a running time of 0.3754 seconds per batch. For the NF-UNSW-NB15-V2 dataset, the running time was slightly higher, at 0.5112 seconds per batch. These results demonstrate that it is capable of processing network traffic data efficiently, making it suitable for real-time intrusion detection scenarios.

G. Time Complexity

The time complexity analysis of TCG-IDS reveals its efficiency in handling dynamic network traffic data. The TGN Encoder operates with a complexity of $O(N_v + N_e)$, where N_v and N_e represent the number of nodes and edges, respectively, ensuring scalability with network size. The temporal contrasting, asymmetric contrasting, and masked contrasting components contribute complexities of $O(N_v \cdot T \cdot D)$, $O(N_v \cdot \bar{d} \cdot D)$, and $O(N_v \cdot M \cdot D)$ respectively, reflecting their operations' dependence on the number of nodes, temporal steps, and embedding dimensions. The combined complexity, $O(N_v \cdot (T + \bar{d} \cdot D + M \cdot D) + N_e)$, highlights the framework's capacity to efficiently process large-scale and temporally evolving network data, making it suitable for real-time intrusion detection applications. This balance between computational efficiency and robust detection underlines the practical applicability of TCG-IDS in diverse network environments.

V. DISCUSSION

While temporal contrasting aims to distinguish embeddings at nearby and distant time steps, this approach may be less effective in the presence of periodic patterns. For instance, network traffic with regular fluctuations, such as nightly or weekend activities, may share consistent underlying patterns across timestamps. Accounting for such periodicity could better preserve meaningful temporal correlations and refine anomaly detection in dynamic networks. Another limitation is the lack of comparison with existing non-GNN methods. While we focused on graph-based approaches due to their relevance in modeling temporal and spatial dependencies, exploring non-GNN methods could provide valuable insights and broaden the scope of evaluation. Future work should incorporate comparisons with more diverse and relevant techniques to further validate the robustness of TCG-IDS.

The lower performance onNF-UNSW-NB15-v2 dataset compared to NF-CSE-CIC-IDS2018-v2 can be attributed to several factors. First, the NF-UNSW-NB15-v2 dataset includes a broader range of attack types, such as Exploits, Backdoor, and Worms, which present more complex detection challenges compared to the more straightforward DoS and DDoS attacks in NF-CSE-CIC-IDS2018-v2. Additionally, NF-UNSW-NB15-v2 features packet-level details that increase the complexity of the detection task, requiring more sophisticated preprocessing and fine-tuning of the model. Finally, differences in the distribution of attacks and traffic patterns between the two datasets influence the model's ability to generalize effectively, further contributing to the performance gap.

The relatively high FPRs for classes such as DoS (12.61%) and Reconnaissance (8.20%) highlight areas for improvement in the model's detection capability. While the framework shows strong overall performance, these high FPRs are significant and will be addressed in future work. Additionally, although the datasets used in this study are based on older data, they remain representative of many real-world scenarios, covering a wide variety of attack types and network traffic patterns. Nevertheless, as cyber threats continue to evolve, it is crucial to evaluate our framework on newer datasets that reflect recent attack techniques and network behaviors to ensure its continued relevance in modern network security contexts.

We reference several relevant graph-based models; however, we specifically selected graph-based models for comparison due to their direct relevance and proven performance in dynamic graph-based tasks, particularly in network intrusion detection. GraphSAGE, which aggregates features from a node's local neighborhood, and GAT, which uses attention mechanisms to weigh neighbors' importance, are both effective for capturing node relationships in network traffic data. These models are highly suitable for handling large-scale dynamic networks and can offer meaningful insights into detection. Although other models may be applicable, we focused on these two because of their successful application in temporal settings and their ability to capture relevant graph structures and temporal dynamics. In future work, we will extend this comparison to include additional models that could provide further insights.

Detecting SSH-brute force and DoS attacks is critical for strengthening network security, as these attacks are noisy and distinct from human behavior. However, their detection can be complicated by factors such as traffic burstiness and network congestion, which may cause them to resemble legitimate high-volume events, increasing false positive rates. Addressing these challenges requires refining feature selection and optimizing thresholding mechanisms to improve detection accuracy and reduce false positives in these attack classes.

It is important to note that while the goal of temporal contrasting is generally to distinguish between embeddings at nearby and distant time steps, this approach might not always be appropriate in the presence of periodic patterns. For example, network traffic at specific times, such as nightly or weekend traffic, might exhibit regular fluctuations, yet still share underlying patterns across different timestamps. In such cases, rather than strictly minimizing the similarity at distant time steps, it may be more beneficial to account for periodicity to preserve meaningful temporal correlations, thus refining the detection of anomalous behaviors in dynamic networks.

The architecture of the proposed framework was determined through a systematic tuning process, including grid search and

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ablation studies, to optimize hyperparameters and component selection. This iterative approach ensured that the model configuration effectively balances performance and computational efficiency, tailored to the characteristics of the datasets used in the study. Future work could explore automated architecture search methods to further refine the framework's design.

VI. Related Work

A. Deep Learning-Based IDS

Deep learning techniques, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have significantly improved network IDS performance. CNNs excel at capturing spatial features in traffic data, while RNNs like LSTMs and GRUs effectively model temporal dependencies [24], [25]. For instance, Kitsune [26] uses autoencoders to detect anomalies, and DeepLog [27] leverages LSTMs for sequence modeling in system logs. However, these methods struggle to simultaneously capture spatial and temporal dependencies and often require extensive labeled data. To overcome these challenges, researchers are increasingly turning to graph neural networks (GNNs) to model interactions and learn rich representations of network data for intrusion detection.

B. GNNs and GNN-Based IDS

GNNs [28], [29], [30], [31] have gained significant attention for learning powerful representations of graph-structured data. They are broadly categorized into spectral-based and spatial-based methods. Spectral methods, like GCNs [17], perform convolution in the Fourier domain using the graph Laplacian's eigendecomposition, but they suffer from high computational complexity and limited scalability. In contrast, spatial-based methods, such as GraphSAGE [16] and GATs [15], aggregate information from neighboring nodes directly on the graph structure, offering greater efficiency and scalability, making them suitable for large-scale and dynamic graphs, such as in network intrusion detection. However, challenges like the over-smoothing problem [32] and limited expressiveness in capturing long-range dependencies [33], [34] remain. Recent innovations like Graph Transformers [35] and Graph Contrastive Learning [36] address these issues, enabling more sophisticated applications of GNNs in complex tasks, including intrusion detection.

Existing GNN-based IDSs can be grouped into three categories: (1) methods leveraging Graph Convolutional Networks (GCNs) or Graph Attention Networks (GATs) for static graph representation, (2) approaches incorporating temporal information through dynamic GCNs or recurrent architectures like LSTMs, and (3) hybrid techniques combining GNNs with reinforcement or adversarial learning. GCNs [37] capture local structural information, while GATs [14] use attention mechanisms for expressive node embeddings but lack temporal considerations. To address this, dynamic GCNs [38] and recurrent models [39] integrate spatial and temporal dependencies, improving detection in evolving networks. Advanced methods like reinforcement learning [40] for subgraph selection and adversarial training [41] enhance scalability and robustness, pushing the boundaries of GNN-based IDSs.

VII. Conclusion

This paper presents TCG-IDS, a self-supervised temporal GNN-based model for network intrusion detection. It uses temporal contrasting, asymmetric contrasting, and masked contrasting to capture temporal dynamics and spatial dependencies in network traffic. By modeling traffic as a dynamic graph, it enables real-time detection with minimal false positives and improved novel attack identification. Evaluation on benchmark datasets shows it surpasses existing models, offering a robust solution for modern network security.

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